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Comparing E-quizzes (Trivia maker™) and paper-based tests on cognitive load, usability, and performance outcomes among university students

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Background: Digital educational platforms have gained notable traction as assessment tools alongside traditional methods. These different assessment tools may impact on several educational and psychological variables. This study investigated the effects of two evaluation modalities, namely Electronic-quizzes and PP, on cognitive load, usability perception, and performance outcomes among university students.

Methods: In a quasi-experimental pre/post-test study design with two intact classes was employed. Ninety-four undergraduate students (age: 23 ± 1.6 years) were allocated by class to an Electronic-Quiz group (E-quiz, n = 47; male = 42%; female = 58%) or a PP group (PP, n = 47; male = 53%; female = 47). Before and after 8 weeks, students completed identical performance-based quizzes and evaluated their cognitive load using the Cognitive Load Scale (CLS), comprising three dimensions: intrinsic load (IL), extraneous load (EL), and germane load (GL), as well as their perception of usability System Usability Scale (SUS).

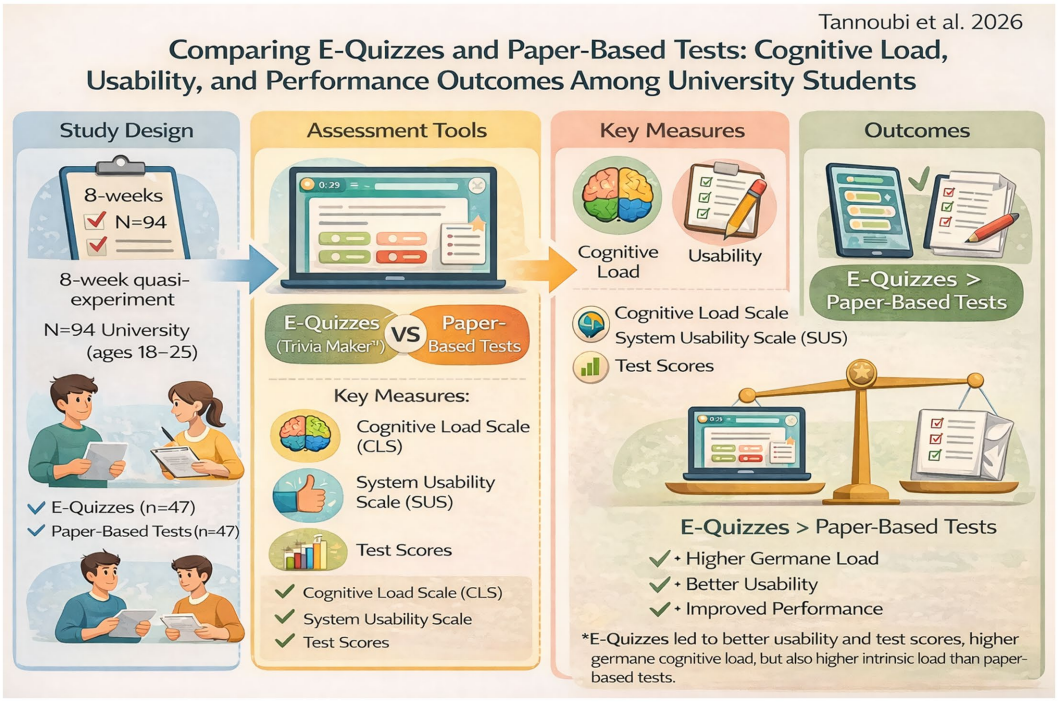
Results: Mixed-model ANOVA revealed significant Time × Group interactions for most primary outcomes. For GL, the E-quiz group showed a large within-group gain (d = 1.40), while the PP group showed negligible change (d = 0.15); between-group post-test difference was medium (d = 0.67, ηp^2 = 0.256). For SUS, the E-quiz group demonstrated large improvement (d = 0.96) compared to PP (d = 0.17), with a medium between-group effect (d = 0.66, ηp^2 = 0.516). For performance, the E-quiz group exhibited a very large within-group gain (d = 1.71), whereas the PP group showed negligible change (d = 0.05); the post-test between-group difference was large (d = 1.82, ηp^2 = 0.623). IL increased only in the E-quiz group (d = 1.15, ηp^2 = 0.203), while both groups showed comparable increases in EL (ηp^2 = 0.030, ns).

Conclusion: In this sample, E-quizzes were associated with larger gains in GL, perceived usability, and academic performance than paper-based tests, and

uniquely elevated perceived intrinsic complexity. These findings align with cognitive load theory predictions regarding the potential role of digital assessment design in supporting germane processing, though the mechanisms remain hypothetical and warrant further investigation. Well-designed digital assessment tools may merit consideration for integration in higher education, contingent on careful design to minimize EL.

KEYWORDS

cognitive load, digital assessment, Electronic quizzes, system usability, undergraduate students



GRAPHICAL ABSTRACT

1 Introduction

Evaluation has an important role in determining the success of learning process. Teachers' evaluation of their students can allow to verify at which level the students accomplish their tasks, remember and understand what they have studied (Hattie and O'Leary, 2025; Nuthall and Alton-Lee, 1995). In the past, the most common tool to assess learners was the use of PP method (Zeiler et al., 2020), which is fair, simple, feasible and low-cost. Currently, educators are increasingly using electronic tests, such as E-quizzes, as education is gradually moving online (Zainuddin et al., 2020). These electronic methods have proven beneficial due to their potential to facilitate quick feedback, allowing the test design's changes, and improve the opportunities for experience-based learning (Pham et al., 2025). However, there is still ongoing debate on whether E-quizzes offer greater efficiency than traditional tests. This test approach can be considered as more effective especially when it is evaluated on how it can affect students' performance, how easy they are to use, and how much mental effort is required from students.

Cognitive load theory (CLT) (Sweller, 1988) offers an extensive framework for understanding the impact of various assessment methods on learners' working memory, cognitive resources, and ultimately their learning performance. CLT delineates three components: intrinsic load (IL; the inherent complexity of the task), extraneous load (EL; associated with the presentation of information) and germane load (GL; which concerns the cognitive effort required for the development and enhancement of mental schemas), (Sweller, 1988, 2011). It is evidenced that the way an assessment is set-up may later change how cognitive resources are spread-out (Greer et al., 2023). Specifically, digital quizzes, for example, could reduce unnecessary activities by clarifying instructions or facilitating navigation (Dermo, 2009; Pengelley et al., 2025; Stödberg, 2012) as well as raising the relevant load by offering timely feedback (Heaslip et al., 2014). However, if the interface is not well designed and learners are accustomed to using it, this can lead to additional demands.

Contrarily, PP tests often are characterized by a cognitive simplicity regarding their presentation, even if they do not allow students to think deeply enough about the subject (Ozuru et al., 2009). Previous research reported that well-structured electronic assessments can enhance cognitive treatment and consequently increase cognitive load and improve learning outcomes (Ihara et al., 2021; Mueller and Oppenheimer, 2014;

Paas and Ayres, 2014). This approach highlights the importance of understanding how testing modality affects learners' cognitive load profiles. However, educators may be unable to benefit from them if they lack digital literacy or experience difficulties with technology.

A further important factor to be considered is ease and enjoyment when using an assessment tool, referred to as system usability. Beyond cognitive load, the perceived usability of an assessment tool represents another critical factor that can influence both cognitive demands and user acceptance (Bangor et al., 2008). According to CLT, a well designed, usable interface should theorically minimize EL—the mental effort spent on navigating the assessment format itself—therby freeing cognitive resources for germane processing and schema construction (Sweller et al., 2019). Enhancing usability—which is measured using the SUS (Brooke, 1996)—can significantly decrease unnecessary cognitive load, by preventing distractions and unjustified frustrations (Carreon, 2025; Clarke et al., 2020; Senthilrajah and Ahangama, 2025). Furthermore, CLT indicates that the total level of cognitive load is less significant than its allocation across intrinsic, extrinsic, and germane dimensions, with GL being especially advantageous for learning (Sweller et al., 2019). E-quizzes (e.g., Trivia maker™) are generally easy to use because they are quick and enjoyable; however, previous research reported mixed results their effectiveness (Enders et al., 2021; Nguyen and McDaniel, 2015; Sanchez et al., 2020). Furthermore, the design of the interface, the user's level of experience, and how easy the use of technology altogether affect usability (Zainuddin et al., 2020). Moreover, PP tests could be regarded as straightforward because learners are not expected to know how to operate a computer (Khan et al., 2021). But they lack features that evolve and adapt to make digital platforms easier to use (Perry et al., 2022; Weirich et al., 2024).

It is important to highlight that the effectiveness of an assessment modality can be interpreted based on its ability to generate reliable proof of learning (Padilla et al., 2023). Previous studies comparing digital and paper-based exams have not yielded sufficient evidences (Fontaine et al., 2021; Li and Yan, 2024). A plethora of research demonstrates that E-quizzes outperform traditional quizzes by providing prompt feedback and reducing students' test-related anxiety (Cao and Han, 2024; Stowell and Bennett, 2010). Other studies have found that E-quizzes had little or no effect on performance as measured by exam scores, and that the design of the task and the characteristics of the learner moderated their effects (Hunsu et al., 2016; Stowell and Bennett, 2010). As well, studies suggest the use of both cognitive and usability criteria to evaluate assessments, instead of just one of them (Kushniruk and Patel, 2004; Polk et al., 2025; Silva et al., 2021). While digital assessment has gained popularity in last few years, to our current knowledge, there has not been a paucity of research that integrally examines the effects of assessments modality on the tripartite construct of cognitive load (IL, EL, GL), usability perceptions, and direct performance outcomes within a single experimental design. Additionally, as digital exams become more widespread in higher education and professional training, it is essential to recognize how they differ from PP tests. Thus, within a quasi-experimental design, this study aims to fill this gap by directly comparing E-quizzes to paper exams in terms of their impact on cognitive load (IL, EL, GL), usability perceptions as measured by the SUS, and a performance indicator. We hypothesized that:

H1: The E-quiz group will show a greater improvement in GL compared to the PP group after the intervention, reflecting deeper cognitive engagement. We posit that the structure, time-bound digital interface may foster greater focus and engagement with the material, directing cognitive resources toward schema construction, even the absence of immediate feedback.

H2: The E-quiz group will report higher system usability scores post-intervention compared to the PP group. We hypothesize that the digital platform's intuitive interface and streamlined interaction will reduce interface-related friction, leading to a more positive perception of usability.

H3: The E-quiz group will achieve better improvement in performance scores compared to the PP group. We posit that this advantage will be mediated by the combined effects of increased germane cognitive engagement (H1) and reduced extraneous demands through higher usability (H2), facilitating more effective learning and retrieval.

2 Materials and methods

2.1 Study design

This study employed a quasi-experimental pre-test/post-test design using two parallel groups. Due to the practical constraints of administering different assessment modalities within intact university classes without disrupting the curriculum, participants were allocated at the class level rather than individually randomized. While this approach enhances ecological validity in a classroom context, it necessarily reduces causal inferences compared with individual level randomization. To mitigate potential confounding from class-level allocation, we ensured instructional materials and teaching were identical across groups, verified baseline equivalence on all primary measures, and controlled for prior digital literacy.

2.2 Participants

The study was carried out with undergraduate students at the High Institute of Sport and Physical Education of Gafsa (ISSEP-G), University of Gafsa, Tunisia. All participants provided written informed consent. Confidentiality and anonymity were ensured, and no sensitive identifying was collected. Ethical approval for the study was obtained from the local ethics committee affiliated with the High Institute of Sport and Physical Education of Gafsa, University of Gafsa, Tunisia, under reference number (ISSEP-GAFSA/EC/2024/091), dated September 23, 2024. In addition, the study protocols complied with the latest ethical standards set out in the Declaration of Helsinki (World Medical Association, 2025).

A priori power analysis was conducted using G*Power (Version 3.1.9.7; Heinrich-Heine-Universität Düsseldorf, Germany) for the *F*-test family (ANOVA: repeated measures, between–within interaction) (Langenberg et al., 2023). Assuming two groups and two measures (before/after), an average effect size of $f = 0.25$, $\alpha = 0.05$, a correlation between repeated measures of $r = 0.50$, and a non-sphericity correction of $\varepsilon = 1$, the analysis indicated a minimum total sample size of $N = 34$ to achieve a power of 80%. The effect size ($f = 0.25$) was selected based on a conservative estimate from prior educational technology studies comparing digital and traditional methods (Hunsu et al., 2016).

Baseline equivalence between the two intact classes was examined prior to the intervention. In addition to age and sex distribution, independent samples t-tests were conducted on all primary outcome variables at pre-test, including IL, EL, GL, system usability, and performance scores. No statistically significant between-group differences were observed at baseline (all $p > 0.10$), indicating comparable starting levels

across outcome domains. Although formal indicators of prior academic achievement (e.g., grade point average) were not collected, pre-test performance scores served as a proximal indicator of baseline academic ability. Furthermore, participants completed a digital literacy survey prior to the intervention, which revealed no significant between-group differences (t (92) = 0.49, p = 0.62), suggesting comparable prior experience with educational technologies. Together, these results support baseline comparability between classes despite class-level allocation.

Two participants were excluded from the initial pool: one from the E-quiz group due to unresolved technical issues preventing quiz completion, and one from the PP group who had prior formal training on the course content. Ninety-four students were therefore selected from two entire classes during regular school hours. Cluster (class-based) allocation was used to assign assessment modalities, resulting in pragmatic quasi-experimental design. One class was assessed using the PP modality (n = 47, male = 53%; female = 47%; age: 23 ± 1.6 years), and the other class received the E-quiz condition (n = 47, male = 42%; female = 58%; age: 23 ± 1.6 years) after receiving the same lessons and learning activities during 8 week intervention period. Further descriptions on the study design and participants are presented in the CONSORT diagram (Figure 1).

2.3 Procedures and timeline description

2.3.1 Introduction and instructions (around 5 min)

The procedures and timelines are presented in Figure 2. At the beginning of the experimentation, the researcher briefly explained to students in the classroom the tasks and gave out directions. The E-quiz group used their phones or tablets to scan a QR code to join the Trivia maker™ platform (see Appendix A). The PP group got printed answer sheets. Discussions about study hypotheses were not allowed between students to avoid impacts of expectation. Both groups were tutored by the same instructor using identical lesson materials. Participants were blinded to the study's specific hypotheses. Additionally, to prevent contamination, the two classes were physically separated during the intervention and post-test phases, and students were instructed not to discuss the procedures.

2.3.2 Taking the quiz (around 15 min)

Each group was asked to complete the identical 12 multiple-choice questions, displaying each one for no more than 45 s. A fixed response time of 45 s per question was imposed to standardize pacing across participants and prevent extended reflection that could differentially advantage one assessment modality. This timing constraint was designed to approximate time-pressured assessment context commonly used in higher education (e.g., in-class quizzes and standardized examinations). Nevertheless, such fixed pacing likely increases cognitive demands across both conditions and may partly explain the observed post-test increase in EL in both groups, as well as the selective increase in IL observed in the E-quiz condition. No immediate feedback was provided during quiz administration for either group. With Trivia maker™ platform, questions were visualized sequentially on screen (see Appendix B).

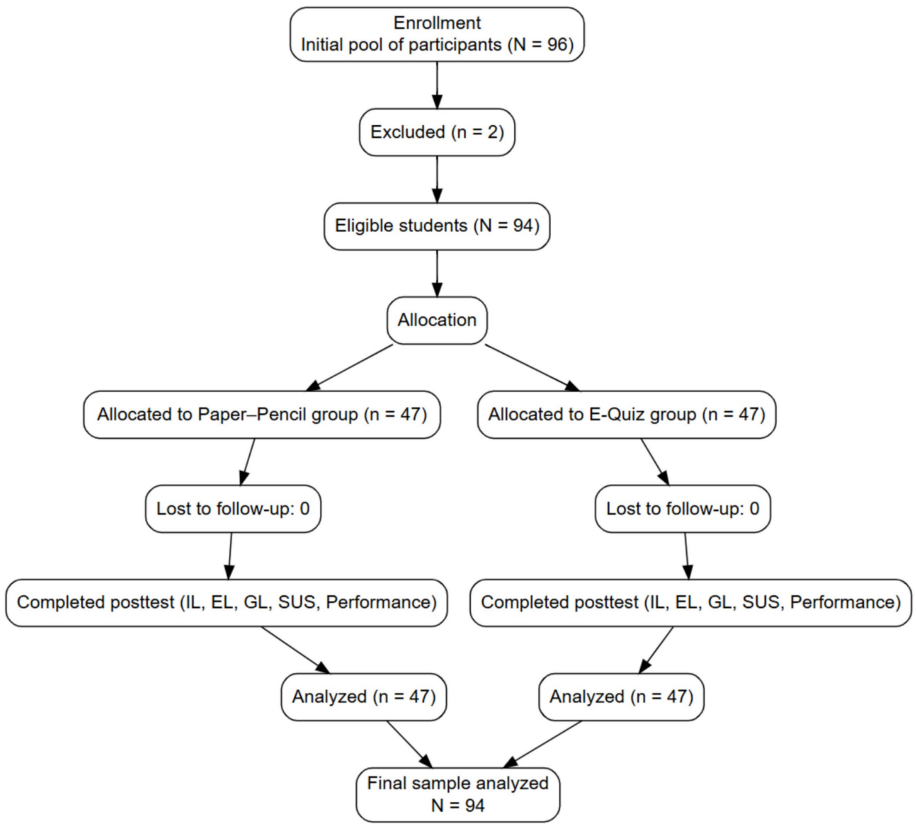


FIGURE 1
CONSORT diagram of the study design.

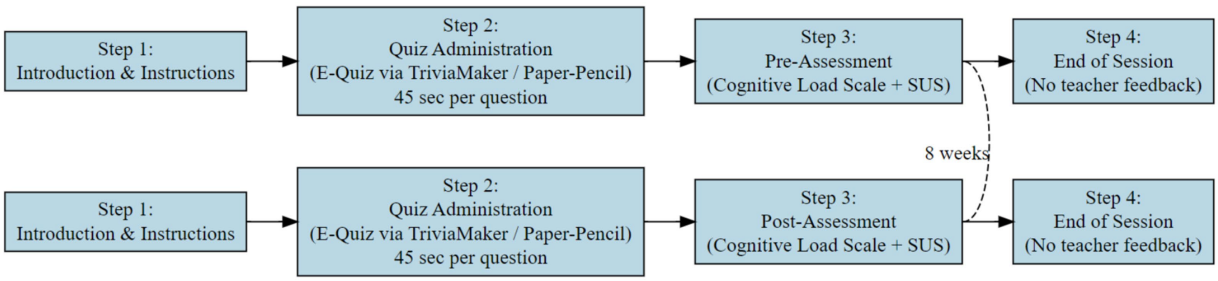


FIGURE 2
Schematic timeline of the study procedures.

The platform was configured to not display correct answers or scores until after the post-test questionnaires were completed, ensuring equivalent feedback timing across both groups. This design ensured that the only difference between groups was the modality (digital vs. paper), not the presence or timing of feedback.

2.3.3 Scales after the pre and post-tests (around 10 min)

After the quiz, participants filled out the Cognitive Load Scale (Leppink et al., 2013) and the SUS (Brooke, 1996). We administered the questionnaires in written form to eliminate any potential communication discrepancies.

2.3.4 Ending of the session (around 2-3 min)

Students resumed their regular activities in class. Consistent with no-feedback protocol, no performance feedback or discussion was provided at any point during the session. Thus, their cognitive load, usability perceptions, and performance scores stayed the same.

2.4 Measures

2.4.1 Cognitive Load Scale

The Cognitive Load Scale (Leppink et al., 2013) was adapted for the quiz context. It comprises three subscales: intrinsic load (IL, 3 items, e.g., "The topics covered in the quiz-based assessment were very complex"), extraneous load (EL, 3 items, e.g., "The instructions and explanations during the quiz-based assessment were very unclear"), and germane load (GL, 4 items, e.g., "The quiz-based assessment really enhanced my understanding of the terminologies covered"). Higher scores in each of the subscales indicate a higher cognitive load. It consists of a 10-point Likert scale, where 1 meant "strongly disagree" and 10 meant "strongly agree." The scale has been shown to be both valid and reliable in different educational settings (Timothy et al., 2023). The Cronbach's α values for the three subscales reported from the original version were between 0.75 and 0.85 (Klepsch et al., 2017; Leppink et al., 2013). In the present sample, internal consistency was acceptable at both pre-test (Cronbach's α: IL = 0.78, EL = 0.76, GL = 0.81) and post-test (IL = 0.80, EL = 0.77, GL = 0.83). Item referencing "problem-based learning (PBL)" were reworded to "quiz-based assessment" to maintain face validity while preserving the original scale structure. In the present study, the intraclass correlation coefficients (ICC) for pre-test and post-test scores were calculated. The ICCs were computed using a two-way random effects model, absolute agreement, single measures [$ICC_{(2,1)} = 0.87$].

2.4.2 System Usability Scale

The SUS (Brooke, 1996) was used to evaluate perceived ease of use of the assessment method. Although originally designed for digital interfaces, prior research supports its adaptation for comparing usability perceptions across digital and non-digital systems when framed generically (Bangor et al., 2008; Lewis, 2018). Participants in both groups were instructed to evaluate "the assessment method you just completed" scores for negatively worded items were reversed, and the sum of all items (range 0–40) was multiplied by 2.5 to yield a total score from 0 to 100, with higher scores indicating better perceived usability. The SUS has proven to be an effective tool for measuring the usability of different systems, with a Cronbach's alpha coefficient ranging from 0.85 to 0.92 (Bangor et al., 2008). The ICC values for the pre-test and post-test were calculated and were 0.89.

2.4.3 Performance scores

The students took a standardized test with 12 multiple-choice questions (see Appendix A). Each question reflects an important idea that was covered throughout the course. The scores ranged from 0 to 12, with higher scores indicating better performance. Two experts checked all the exam questions to ensure that they were clear, relevant, and covered all the course's stated learning goals. Both groups took the same test in the same classroom to ensure comparability between assessment modalities. To ensure scoring consistency, a random subset of answer sheets (20%) was independently scored by a second rater. Inter-rater reliability, calculated via ICC, was excellent (ICC = 0.96).

2.4.4 Statistical analyses

Data were input into *Microsoft Excel* (Microsoft Corporation, 2025, Redmond, WA, United States) and all statistical analyses were performed using JASP software, version 0.95.0.0 (JASP Team, 2025). The descriptive statistics of all variables, including means, standard deviations, and medians were examined. Data normality was assessed using Shapiro-Wilk's tests (Shapiro and Wilk, 1965). As shown in Table 1, these tests indicated some deviations from normality, particularly for performance scores at specific time points. However, inspection of skewness and kurtosis values indicated that all distributions fell within the commonly accepted range of −2 to +2, which is widely regarded as indicative of approximate normality in applied behavioral and educational research (George, 2011; Hopkins and Weeks, 1990).

Moreover, each experimental cell included more than 30 participants, satisfying conditions under which the central limit theorem supports the robustness of parametric tests to moderate violations of normality. Simulation and methodological studies have demonstrated that mixed-model ANOVA remains robust under such conditions, with minimal inflation of Type I error rates, particularly when group sizes are balanced (Schmider et al., 2010). To further ensure the validity of the findings, confirmatory non-parametric analyses using aligned rank transform procedures were conducted for all primary outcomes. These analyses yielded substantively identical interaction effects, supporting the robustness of the reported parametric results.

Participants were allocated at the class level, resulting in two intact clusters. Although multilevel mixed-effects models are often recommended for clustered data, the inclusion of class as a random effect was not feasible in the present study due to the very small number of clusters ($k = 2$). Methodological literature indicates that random-effects estimation with fewer than five clusters yields unstable variance components and unreliable fixed-effect estimates (McNeish, 2019; McNeish and Kelley, 2019). Consequently, a class-level random factor was not modeled. To mitigate this limitation, we verified baseline equivalence across groups on all primary outcomes, standardized instructional delivery and assessment procedures, and emphasized interaction patterns and effect sizes rather than statistical significance alone. The analyses therefore estimate associations between assessment modality and outcomes rather than definitive causal effects.

A mixed-model analysis of variance (ANOVA) [Time (pre and post) × group (PP and E-quiz)] was then performed to examine the main effect of time and group, as well as their interaction effects on cognitive load dimensions (IL, EL, and GL), system usability, and performance scores. Partial eta squared (ηp^2) effect size values were calculated and classified as 0.01 = small, 0.06 = medium, 0.14 = large and Cohen's d (small = 0.20, medium = 0.50, large = 0.80) was used to interpret the magnitude of differences (Cohen, 2013). The R Studio version 2025.05.0 for Windows (R Core Team, 2024) was used to produce the figures. The *ggplot2*, *ggtext*, and *showtext* packages were used (Qiu, 2024; Wickham, 2016; Wilke, 2022). The significance level for all analyses was $p < 0.05$.

3 Results

3.1 Descriptive analysis

Table 1 presents descriptive statistics for all outcomes according to groups and time of measurement. Normality assessment revealed some deviations in performance scores at pre-test (Shapiro-Wilk $p = 0.01$–0.06), though skewness and kurtosis values were within acceptable bounds [−2, +2] for all variables. Given adequate sample sizes and ANOVA's robustness to such deviations, parametric analyses were maintained.

3.2 Mixed ANOVA results for cognitive load, usability, and performance measures

3.2.1 Intrinsic load

A significant Time × Group interaction was found for IL, $F_{(1,92)} = 23.45$, $p < 0.001$, $\eta p^2 = 0.203$ [95% CI: 0.110, 0.312]. Simple effects analysis indicated that the E-quiz group's IL increased

TABLE 1 Descriptive statistics and normality.

Variable	Group	Time	*n*	M	SD	Median	Skewness	Kurtosis	*W*	*p* (Shapiro-Wilk)
IL	PP	Pre	47	13.17	5.25	13.50	−0.32	−1.07	0.977	0.48
		Post	47	12.24	4.70	13.00	0.84	−0.27	0.959	0.11
	E-quiz	Pre	47	11.83	5.43	11.50	0.65	−0.52	0.973	0.33
		Post	47	18.15	5.60	18.00	0.99	0.62	0.981	0.60
EL	PP	Pre	47	10.21	5.58	10.00	0.64	−0.72	0.952	0.06
		Post	47	15.80	5.56	16.00	0.41	−0.76	0.975	0.41
	E-quiz	Pre	47	10.54	4.96	10.00	0.79	−0.12	0.976	0.42
		Post	47	14.37	5.41	14.00	1.58	1.68	0.983	0.70
GL	PP	Pre	47	21.33	8.02	22.00	−0.51	−0.68	0.988	0.90
		Post	47	22.52	7.67	24.50	−0.90	0.12	0.959	0.10
	E-quiz	Pre	47	19.02	5.00	19.50	−1.23	1.08	0.985	0.81
		Post	47	27.31	6.70	27.50	−1.53	1.79	0.985	0.77
SUS	PP	Pre	47	71.40	8.55	72.10	−0.33	−0.48	0.981	0.65
		Post	47	72.99	10.26	75.20	−0.60	−0.12	0.970	0.27
	E-quiz	Pre	47	70.49	9.11	68.55	−0.51	0.06	0.960	0.10
		Post	47	79.50	9.60	80.25	−0.78	1.16	0.978	0.52
Performance	PP	Pre	47	5.85	1.98	6.00	0.24	−0.54	0.934	0.01
		Post	47	5.96	2.28	6.00	−0.19	0.18	0.968	0.23
	E-quiz	Pre	47	6.33	2.15	6.00	−0.11	−0.70	0.955	0.06
		Post	47	9.35	1.25	9.00	−0.40	0.21	0.932	0.01

M, Mean; SD, standard deviation; IL, intrinsic load; EL, extraneous load; GL, Germane load; SUS, System Usability Scale; PP, paper pencil; E-quiz, Electronic-quiz.

significantly pre- to post-test [$p < 0.001$, $d = 1.15$ (95% CI: 0.75, 1.55)], whereas the PP group showed no significant change [$p = 0.32$, $d = -0.19$ (95% CI: −0.59, 0.22)]. The post-test between-group difference in IL represented a large effect [$d = 1.15$ (95% CI: 0.75, 1.55)]. There were no significant main effects of time, $F_{(1,92)} = 0.15$, $p = 0.702$, $\eta p^2 = 0.002$ [95% CI: 0.000, 0.029], or group, $F_{(1,92)} = 1.12$, $p = 0.293$, $\eta p^2 = 0.012$ [95% CI: 0.000, 0.051] (see Table 2 and Figure 3).

3.2.2 Extraneous load

No significant Time × group interaction was found for EL [$F_{(1,92)} = 2.87$, $p = 0.094$, $\eta p^2 = 0.030$ (95% CI: 0.000, 0.065)], indicating both groups reported comparable increases in EL over time. A significant main effect of time [$F_{(1,92)} = 15.00$, $p = 0.001$, $\eta p^2 = 0.140$ (95% CI: 0.060, 0.236)] confirmed this overall increase. There was no significant main effect of group [$F_{(1,92)} = 0.07$, $p = 0.75$, $\eta p^2 < 0.001$ (95% CI: 0.000, 0.028)]. Within-group effect sizes showed that both the E-quiz [$d = 0.73$ (95% CI: 0.43, 1.04)] and PP [$d = 1.00$ (95% CI: 0.68, 1.32)] groups experienced medium increases in EL, with no meaningful post-test between-group difference ($d = -0.26$) (see Table 2 and Figure 3).

3.2.3 Germane load

A significant Time × Group interaction was found for GL, $F_{(1,92)} = 31.67$, $p < 0.001$, $\eta p^2 = 0.256$ [95% CI: 0.167, 0.345], indicating differential change patterns between groups. Simple effects analysis showed that the E-quiz group's GL increased significantly more than the PP group's (both $p < 0.001$). Effect size analyses revealed that the

TABLE 2 Cognitive load, usability, and performance measures before and after intervention in PP and E-quiz groups.

Measure	Effect	*F*	df	*p*	ηp^2 [95% CI]	Cohen's *d*
IL	Time	0.15	1.92	0.702	0.002 [0.000, 0.029]	—
	Group	1.12	1.92	0.293	0.012 [0.000, 0.051]	—
	Time × group	23.45	1.92	<0.001	0.203 [0.110, 0.312]	E-quiz: 1.15[a]
						PP: −0.19[a]
						Post Δ: 1.15[b]
EL	Time	15.00	1.92	<0.001	0.140 [0.060, 0.236]	—
	Group	0.07	1.92	0.75	0.001 [0.000, 0.028]	—
	Time × group	2.87	1.92	0.094	0.030 [0.000, 0.065]	E-quiz: 0.73[a]
						PP: 1.00[a]
						Post Δ: −0.26[b]
GL	Time	25.40	1.92	<0.001	0.23 [0.130, 0.340]	—
	Group	0.38	1.92	0.539	0.004 [0.000, 0.035]	—
	Time × group	31.67	1.92	<0.001	0.256 [0.167, 0.345]	E-quiz: 1.40[a]**
						PP: 0.15[a]
						Post Δ: 0.67[b]
SUS	Time	4.85	1.92	0.030	0.050 [0.010, 0.108]	—
	Group	0.84	1.92	0.362	0.009 [0.000, 0.043]	—
	Time × group	98.24	1.92	<0.001	0.516 [0.395, 0.620]	E-quiz: 0.96[a]**
						PP: 0.17[a]
						Post Δ: 0.66[b]
Performance	Time	97.52	1.92	<0.001	0.515 [0.393, 0.619]	—
	Group	61.83	1.92	<0.001	0.402 [0.282, 0.509]	—
	Time × group	152.36	1.92	<0.001	0.623 [0.507, 0.724]	E-quiz: 1.71[a]**
						PP: 0.05[a]
						Post Δ: 1.82[b]

[a]Cohen's *d* for within-group pre–post changes (interpretation: $d < 0.20$ = negligible; 0.20–0.50 = small; 0.50–0.80 = medium; >0.80 = large).
[b]Cohen's *d* for post-test between-group differences (E-quiz vs. PP).
df, degrees of freedom; ηp^2, partial eta-squared. *, $p < 0.05$; **, large effect size ($d > 0.80$). Time, pre vs. post; Group, PP vs. E-quiz; E-quiz, electronic-quiz; PP, paper-pencil.

E-quiz group demonstrated a large within-group pre-to-post increase [$d = 1.40$ (95% CI: 1.00, 1.81)], whereas the PP group showed negligible change [$d = 0.15$ (95% CI: −0.25, 0.56)]. The between-group post-test difference represented a medium effect [$d = 0.67$ (95% CI: 0.26, 1.07)], indicating a substantively meaningful advantage for the E-quiz format. Significant main effects were found for time, $F_{(1,92)} = 25.40$, $p < 0.001$, $\eta p^2 = 0.230$ [95% CI: 0.130, 0.340], but not group, $F_{(1,92)} = 0.38$, $p = 0.539$, $\eta p^2 = 0.004$ [95% CI: 0.000, 0.035] (see Table 2 and Figure 3).

3.2.4 System Usability Scale

A significant Time × Group interaction was found for SUS, $F_{(1,92)} = 98.24$, $p < 0.001$, $\eta p^2 = 0.516$ [95% CI: 0.395, 0.620], demonstrating differential improvement in usability between groups. Simple effects analysis revealed that the E-quiz group showed a significantly greater increase in usability scores than the PP group (both $p < 0.001$). Within-group effect sizes indicated that the E-quiz group experienced a large pre-to-post improvement [$d = 0.96$ (95% CI: 0.57, 1.35)], while the PP group showed negligible change [$d = 0.17$ (95% CI: −0.24, 0.58)]. The post-test between-group difference represented a medium effect [$d = 0.66$ (95% CI: 0.25, 1.07)]. A significant main effect of time was found, $F_{(1,92)} = 4.85$, $p = 0.030$, $\eta p^2 = 0.050$ [95% CI: 0.010, 0.108], but not group, $F_{(1,92)} = 0.84$, $p = 0.362$, $\eta p^2 = 0.009$ [95% CI: 0.000, 0.043] (see Table 2 and Figure 3).

3.2.5 Performance

A significant Time × Group interaction was found for performance, $F_{(1,92)} = 152.36$, $p < 0.001$, $\eta p^2 = 0.623$ [95% CI: 0.507, 0.724], demonstrating that the extent of improvement over time differed between groups. Simple effects analysis showed that the E-quiz group exhibited significantly greater performance gains than the PP group (both $p < 0.001$). Effect size analyses revealed that the E-quiz group demonstrated a very large within-group pre-to-post improvement [$d = 1.71$ (95% CI: 1.29, 2.13)], whereas the PP group showed a negligible effect [$d = 0.05$ (95% CI: −0.36, 0.46)]. The between-group post-test difference was also large [$d = 1.82$ (95% CI: 1.39, 2.25)]. Significant main effects were also found for time, $F_{(1,92)} = 97.52$, $p < 0.001$, $\eta p^2 = 0.515$ [95% CI: 0.393, 0.619], and group, $F_{(1,92)} = 61.83$, $p < 0.001$, $\eta p^2 = 0.402$ [95% CI: 0.282, 0.509] (see Table 2 and Figure 3).

A visual summary of the effect sizes for the Time × Group interactions is presented in Figure 3.

4 Discussion

The present study assessed the effects of using two different evaluation modalities (i.e., PP and E-quizzes) on cognitive load dimensions (i.e., IL, EL, and GL), usability perception, and performance within university students. While all three hypotheses were supported, the results revealed nuances, modality-specific effects. Significant Time × Group interactions were the central finding: the E-quiz demonstrated substantially greater improvements than the PP group in GL, usability, and performance, confirming all three hypotheses (H1, H2, H3) with large effect sizes (see Table 2 and Figure 3). A significant interaction was also found for IL, indicating that the E-quiz group perceived the test as more complex over time, while the PP group did not. No significant interaction was observed for EL, suggesting that both formats imposed similar levels of unnecessary cognitive demand.

The significant Time × Group interaction for GL supports H1 and aligns with CLT emphasis on GL as facilitative of schema construction (Sweller, 1988, 2011; Sweller et al., 2019). Unlike earlier interpretations, the interaction confirms that the rate of improvement in GL was greater

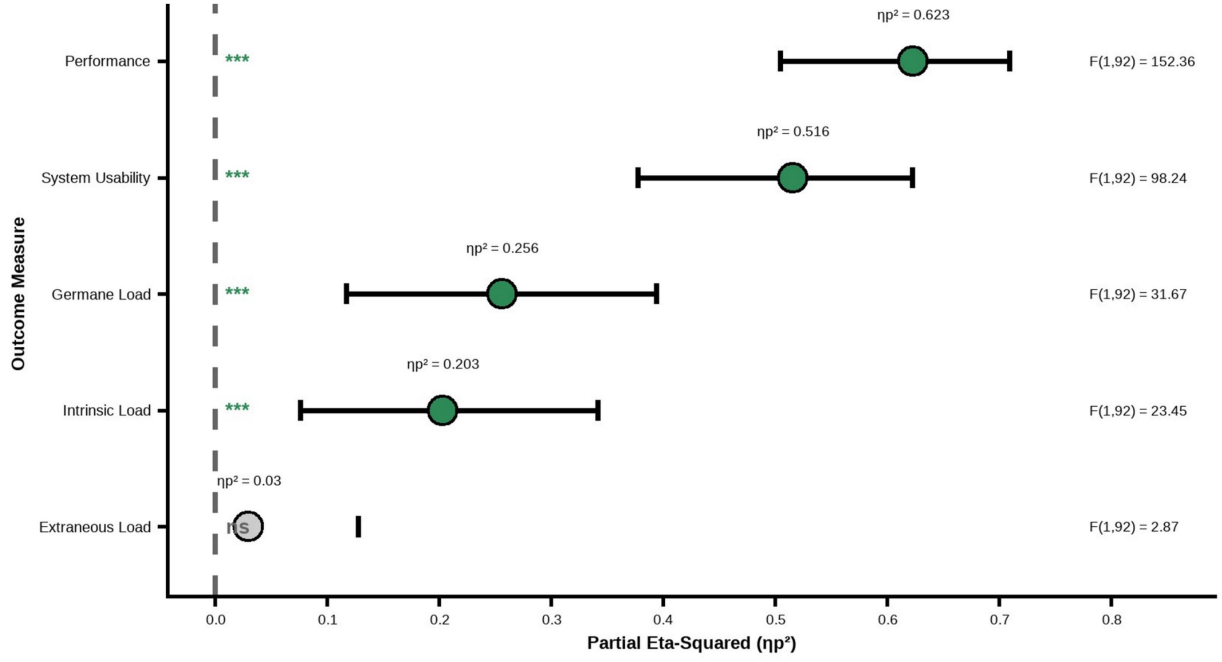


FIGURE 3
Forest plot of partial eta-squared (ηp^2) effect sizes for significant time × group interaction effects across outcome measures. ***$p < 0.001$, ns, not significant.

for the E-quiz group, indicating that the digital format more effectively fostered the cognitive engagement necessary for deeper learning. This finding, supported by a large effect size ($\eta p^2 = 0.256$), highlights the importance of distinguished not only between the final level of cognitive engagement and the rate of its development (Paas and Ayres, 2014), but also underscores how digital tools can differentially promote germane processing over time. An alternative explanation for the observed advantages of the E-quiz condition relates to novelty and gamification effects. Although baseline digital literacy did not differ between groups, the game-like and interactive nature of the Trivia maker™ platform may have temporarily enhanced motivation, attention, or test engagement. Such novelty effects are well documented in educational technology research and may partially mediate performance and cognitive engagement outcomes independently of cognitive load mechanisms.

The significant interaction for IL suggests that the assessment modality influenced students' perception of the task's inherent difficulty. The increase in IL reported by the E-quiz group may stem from the added cognitive layer introduced by the digital interface, such as navigating on-screen questions, adapting to platform pacing, or managing interactive elements, factors absent in the static paper-based format. This increase may be understood as the digital interface adding layer of element interactivity—managing navigation, on screen presentation, and platform pacing—which, according to CLT, can increase the intrinsic complexity of a task as perceived by the learner (Sweller, 2011). This align with CLT, which posits that the manner of information presentation (extraneous factor) can shape the perceived complexity (IL) of the task itself (Sweller, 2011; Sweller et al., 2019). In contrast, the stability of IL in the PP group suggests that the traditional format did not alter the perceived core difficulty of the quiz content.

No significant Time × Group interaction was observed for EL, indicating that the pattern of change over time did not differ between assessment modalities. However, a significant main effect of time revealed that EL increased from pre- to post-test in both groups. This suggests that both the electronic and paper-based formats imposed comparable increases in unnecessary cognitive demands. A plausible explanation for this uniform increase is the fixed 45 s response limit per item, which likely introduced time pressure irrespective of modality. Time pressure irrespective of modality. Time pressure is a well-established source of extraneous cognitive load and may override potential modality-specific advantages in interface design that could otherwise reduce unnecessary cognitive processing under more self-paced assessment conditions.

Regarding usability, the significant interaction confirms H2 and indicates that the E-quiz group perceived the digital platform as increasingly usable over time, with a significant improvement in SUS scores post-intervention. This aligns with prior research highlighting that digital assessment tools are valued for their accessibility, efficiency, and reduced administrative burden (Aristeidou et al., 2024; Washburn et al., 2017). Usability is not merely a superficial feature; it directly influences cognitive load by reducing interface-related friction, thereby freeing cognitive resources for learning tasks (Chu et al., 2019; Vlachogianni and Tselios, 2022). The high SUS scores in the E-quiz group likely contributed to the platform's effectiveness by minimizing extraneous demands and supporting sustained engagement. While the correlational nature of this relationship within our design precludes definitive causal conclusions, the alignment of high usability scores with superior performance in the E-quiz group supports the theoretical premise that a reduction in interface-related friction can liberate cognitive resources for learning.

The superior performance outcomes in the E-quiz group strongly support H3 with a large effect size ($\eta p^2 = 0.623$) underscoring the pedagogical potential and are consistent with a grow body of evidence on technology-enhanced learning (Hunsu et al., 2016; Kennette and McGuckin, 2018; Stowell and Bennett, 2010). The magnitude of the improvement aligns with research on retrieval-based practice, which shows that repeated, low-stakes testing can significantly boost long-term retention and transfer (Buchin and Mulligan, 2023; Karpicke and Aue, 2015; Karpicke et al., 2016; Karpicke and Grimaldi, 2012). The digital format likely facilitated this through immediate item presentation, structured pacing, and an engaging interface. While the PP group also showed a modest improvement, the differential gain in the E-quiz group underscores the pedagogical potential of well-designed digital assessments.

In summary, this study demonstrated that E-quizzed outperform paper-based tests in promoting GL, usability, and academic performance. The digital format also uniquely influenced the perception of task complexity, likely due to interface-related cognitive layers. These findings reinforce CLT and highlight the importance of usability in digital learning tools. Future studies should explore the long-term retention of these benefits, the role of individual differences, and ways to further minimize EL in digital assessment environments. Because both groups received identical instruction over 8 weeks, some improvement in performance likely reflects effects or general maturation. However, the minimal change observed in the PP group compared with the substantial gain in the E-quiz group suggests that practice effects alone cannot account for the observed interaction. Nevertheless, the 12-item multiple-choice format primarily assessed factual knowledge and may not be sensitive to higher-order cognitive processes such as transfer or problem solving.

4.1 Study limitations and future directions

This study has several limitations that should be acknowledged. First, the use of cluster (class-level) allocation with only two intact classes limits internal validity. Because the number of clusters was insufficient to estimate intra-class correlation or include class as a random effect, standard errors may be underestimated and statistical significance potentially inflated. Although large effect sizes were observed for GL, usability, and performance, these findings should be interpreted cautiously and viewed as exploratory rather than confirmatory. Future studies should prioritize individual randomization where feasible or apply matching procedures (e.g., propensity score matching) to better control for pre-existing differences between learners. Second, the fixed 45 s response limit, while standardizing pacing, likely increased EL and IL for all participants and reduces ecological validity compared to self-paced, real world assessments. Third, applying the SUS, originally designed for digital interfaces, to a paper-based test, despite adapted instructions, remains conceptually unconventional. Fourth, the sample was drawn from a single institution and discipline, which may limit generalizability to other academic domains, educational levels, or cultural contexts. Fifth, the absence of delayed post-test precludes conclusions about long-term retention of learning gains. Finally, the reliance on quantitative measures alone limits the depth of understanding regarding the underlying experiences and cognitive processes of participants. Future investigations should control prior experience with digital quizzes, include familiarization phases, or employ crossover design to disentangle modality effects from novelty-driven engagement. Additionally, future research should employ individual randomization where feasible, use self-paced timing to enhance ecological validity, include longitudinal and multisite multidisciplinary samples, and

integrate mixed methods approaches to better elucidate the mechanisms and lasting effects of digital assessment tools.

4.2 Practical and educational implications

From an educational perspective, the results of the present study have several implications. Using E-quizzes in regular classroom lessons can help increase student attention and improve performance outcomes, supporting recommendations for higher education to integrate regular, low-stakes digital assessments. Furthermore, the concurrent improvement in GL and performance suggests that well-designed E-quizzes can foster deeper, schema-based learning rather than superficial memorization. Usability should not be overlooked, as platforms that are intuitive and easy to navigate ensure equitable participation and reduce initial barriers to engagement.

4.3 Implementation considerations

Educators should note that the comparable rise in EL across both groups suggests that merely digitizing assessments is insufficient. To fully realize the benefits observed in GL and performance, digital platforms must be carefully designed and implemented to minimize EL. This includes using intuitive interfaces, providing clear instructions, allowing for practice sessions to build familiarity, and considering flexible time limits to avoid inducing unnecessary time pressure.

5 Conclusion

This study demonstrates that E-quizzes were associated with higher germane cognitive load, improved usability perceptions, and superior performance outcome compared with paper-based tests. The findings further indicate that assessment modality can influence students' perceived intrinsic task complexity, with the electronic format eliciting higher IL over time, likely due to interface-related cognitive demands. No modality-specific differences were observed for EL, suggesting that shared design constraints, particularly imposed time pressure, may play a dominant role in shaping unnecessary cognitive demands across assessment formats. Overall, the observed pattern of results is consistent with predictions derived from CLT, although the underlying mechanisms remain hypothesized rather than directly tested. Taken together, these findings support the careful integration, of well-designed digital assessment tools in higher education, while underscoring the importance of usability, pacing, and learner familiarity in optimizing cognitive engagement and learning outcomes.

Data availability statement

The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation.

Ethics statement

Ethical approval for the study was obtained from the Local Ethics Committee affiliated with the High Institute of Sport and Physical Education of Gafsa, University of Gafsa, Tunisia, under reference number (ISSEP-GAFSA/EC/2024/091), dated September 23, 2024. The studies were conducted in accordance with the local legislation and institutional requirements. The participants provided their written informed consent to participate in this study. Written informed consent was obtained from the individual(s) for the publication of any potentially identifiable images or data included in this article.

Author contributions

AT: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Software, Writing – original draft, Writing – review & editing. IO: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Writing – original draft, Writing – review & editing. TB: Investigation, Methodology, Validation, Visualization, Writing – review & editing. MS-S: Investigation, Methodology, Validation, Visualization, Writing – review & editing. JH: Funding acquisition, Investigation, Methodology, Validation, Visualization, Writing – review & editing. ES: Investigation, Methodology, Validation, Visualization, Writing – review & editing. FA: Investigation, Methodology, Supervision, Validation, Visualization, Writing – review & editing.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

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Appendix A

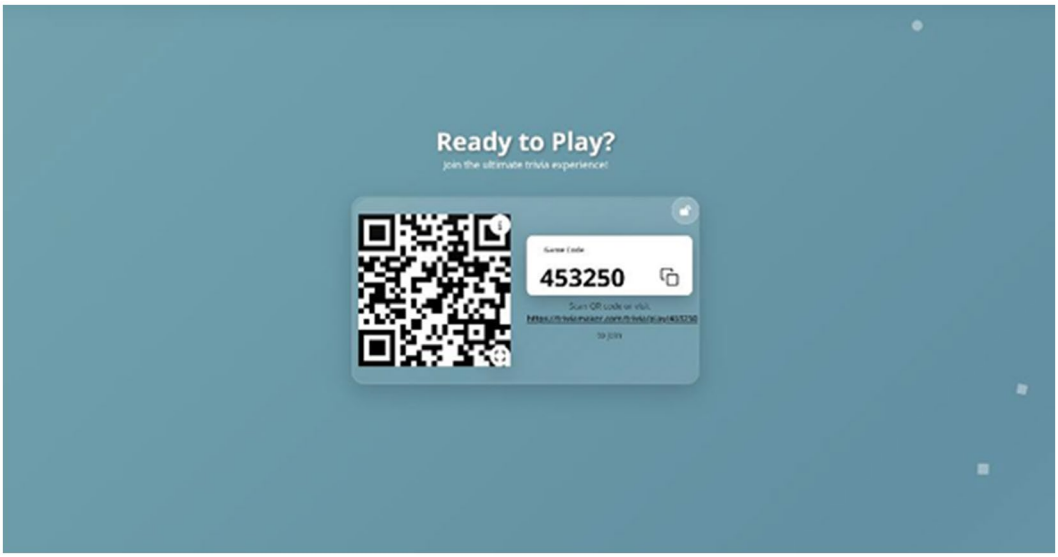


FIGURE A1
QR code to join the E-quiz (Trivia maker™).

Appendix B

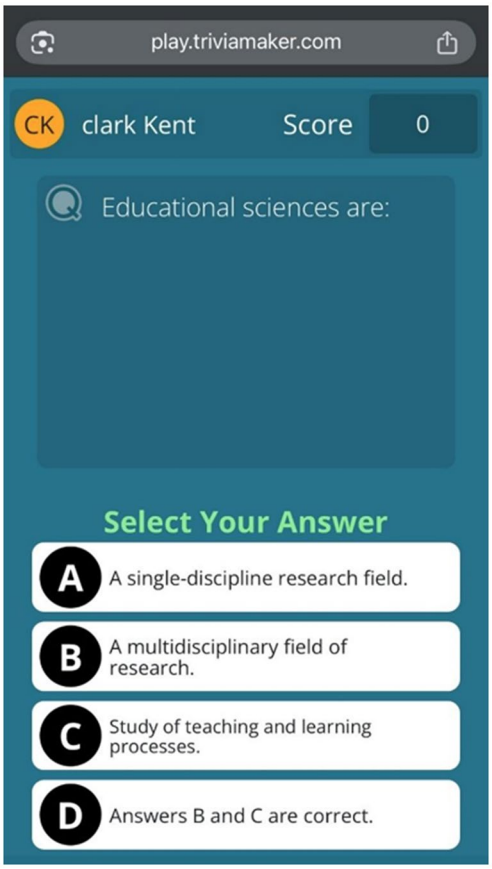


FIGURE B1
Participant interface (smartphones/tablets).